Week 12: Security Analytics and Final Project Launch

Today we finish the analytics unit and begin the final project in a structured, manageable way.

First half

security analytics
metrics as evidence
privacy-aware monitoring

Second half

final project choices
beginner project scope
in-class lab due today

Work mode

individual work
specific topic
clear evidence

Course Map

Where Week 12 fits in the course

Syllabus sequence

Week 11-12: Data Analytics for Security

Week 13: Data Poisoning and Model Evasion

Week 14: Final Project Presentations

Week 15: Final Project Due and Wrap-Up

Why this week matters

Analytics turns security data into evidence.

The final project turns one course question into a focused argument.

This week gives students time to start the final report.

Analytics

Security analytics means finding useful security meaning in data

Input

logs
alerts
asset context
user activity
cloud events

Thinking

filter
count
compare
join context
notice patterns

Output

priority
next step
risk explanation
short analyst note

Workflow

A practical analytics workflow

1. Collect: decide what security data is relevant.
2. Clean: make the data readable and consistent.
3. Enrich: add context about assets, users, or sensitivity.
4. Compare: look for patterns, changes, and outliers.
5. Decide: choose what deserves attention first.
6. Document: write what you found and why it matters.

Metrics are signals, not automatic conclusions

A metric can help

failed logins can suggest attack activity

MFA denials can suggest account pressure

malware alerts can suggest system compromise

sensitive data can raise the priority

A metric can mislead

registration week may increase normal login failures

public systems may create noisy alerts

one high number may not prove compromise

missing context can distort priority

Privacy

Privacy-aware monitoring asks what should be collected, limited, and deleted

Purpose

Why are we collecting this?

What decision will it support?

Who benefits from the monitoring?

Limits

Who can access it?

How long is it kept?

Can it be used for another purpose?

Risk

Could it expose people?

Could it be misread?

Could it become surveillance?

Lab

Today's lab: read a small security metrics table

What you'll do

- open `week12_security_metrics.csv`
- score at least five systems
- choose one system to review first
- write a short analyst note

Skills you'll practice

- reading rows and columns
- separating signal from context
- explaining priority
- adding a privacy boundary

Final Project

The final project is one focused question answered with evidence

The project is not

a giant expert-level investigation

a hacking assignment

a vague opinion paper

a huge app students cannot finish

The project is

one clear question

course concepts applied carefully

evidence from data, sources, or a case

a realistic recommendation

Two Paths

Students choose one path: security analyst code project or MLA research essay

Code path

- small notebook or script
- class CSV or fictional data
- analyst-style findings
- short report

Essay path

- brief MLA research essay
- credible sources
- in-text citations
- Works Cited

Same goal

- clear question
- course concepts
- security and privacy thinking
- clear final report

Code Option

Coding project: build a small beginner analyst task

Good analyst projects

login review notebook
alert triage scorecard
security metrics daily summary
privacy-aware asset inventory
cloud database controls checklist

Minimum code work

load a CSV
filter, group, sort, or score rows
make one summary table
write an analyst-style explanation

Essay Option

Research essay: use MLA to analyze one covered course issue

Good essay topics

- student monitoring software
- data brokers and linked datasets
- facial recognition
- cloud database exposure
- security logging and surveillance

Essay structure

- focused question
- background
- security issue
- privacy or ethics issue
- MLA citations and Works Cited

Topic Scope

Beginner projects work best when they are narrow

Too broad: Is AI dangerous?

Better: How can automated classification create privacy and bias risks?

Too broad: What is hacking?

Better: How can login logs reveal password-spray behavior?

Too broad: Is cloud secure?

Better: What controls reduce cloud database exposure?

Evidence

Good final projects use evidence, not just opinion

Code evidence

- tables
- charts
- counts
- risk scores
- data dictionary

Research evidence

- official docs
- standards
- case reports
- news investigation
- academic source

Course evidence

- authentication
- access control
- databases
- cloud
- privacy
- ethics

Responsibility

Responsible project boundaries

Allowed

- fictional data
- provided class CSVs
- public sources
- official documentation
- safe analysis and critique

Not allowed

- attacking real systems
- scanning websites without permission
- collecting private data from real people
- building malware or phishing tools

Proposal

The in-class lab turns a topic into a plan

Question

What exactly are you investigating?

What system, case, or dataset is involved?

Evidence

What data or sources will you use?

Are they safe and credible?

Plan

What are your next three tasks?

What help do you need?

Timeline

A manageable timeline from today to the final report

- Week 12: choose path, question, data or sources, and complete the in-class lab.
- Before Week 13: collect sources or prepare the dataset.
- Week 13: build the first analysis or draft the essay argument.
- Week 14: continue building the final report.
- Week 15: submit the final report and wrap up.

Work Time

What to do during the second half of class

Finish first

choose analyst code or MLA essay

write one focused question

pick data or sources

name two course concepts

Then start

coding path: open a notebook and load class or fictional data

essay path: find sources and plan MLA citations

both paths: write next three tasks

Day 2 Studio

Day 2 is mostly individual final project lab time

Short class framing

- review the two paths
- check project scope
- answer common questions
- remind students about boundaries

Most of class

- finish the in-class lab
- source or dataset check
- outline building
- first code step or MLA essay claim

End of Week 12

By the end of the week, every student should know what they are making and what evidence they will use.

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